

MICHAEL SHEPARD

Ann Arbor, Michigan

📞 734-596-0203 ✉️ michshep@umich.edu 🔗 [linkedin.com/in/michael-shepard-356288233](https://www.linkedin.com/in/michael-shepard-356288233) 🌐 github.com/MichShep

Education

University of Michigan

Bachelor of Science in Computer Science with Minor in Electrical Engineering

Aug. 2022 – Dec 2026

Ann Arbor, Michigan

Technical Skills

Programming: C, C++, ARM Assembly, Python, Bash, MATLAB

Embedded Systems: Bare-Metal Programming, Interrupts, DMA, Memory-Mapped I/O, Register-Level Programming, Memory-Constrained Systems

Hardware & Architecture: SystemVerilog, Verilog, RISC-V, RTL Design, Superscalar Architecture, Logic Synthesis

Software Engineering: Data Structures & Algorithms, System Design, Multithreading, Client-Server Networking

Tools: Linux, GDB, Git, Make, CMake, Valgrind, Synopsys Design Compiler

Experience

NASA SunRISE Heliophysics Mission — University of Michigan

Jan. 2025 – Present

Software and Data Analysis Co-Lead

Ann Arbor, Michigan

- Co-lead a software and data-analysis subteam of **10+** engineers, coordinating requirements and system integration with electrical and mechanical teams supporting a **20+** antenna ground array.
- Developed and optimized a **multithreaded C++ FX-correlation and beamforming** pipeline for processing ground-array observations, reducing computational latency by **15%**.
- Designed **transient-detection and classification** pipelines and maintained a labeled observational database for identifying solar radio bursts in high-noise data, achieving **84%** detection accuracy.
- Configured, integrated, and troubleshot **ground-array computing hardware** and data-processing workflows to support reliable acquisition and analysis of antenna data on a live NASA mission.

Projects

GBA Bare-Metal Game Engine | C, ARM Assembly (ARM7TDMI), Make, GDB

Aug. 2025 – Present

- Developed a **bare-metal** embedded system in C and ARM assembly for the ARM7TDMI, directly configuring memory-mapped display, interrupt, and DMA registers without an operating system.
- Sustained **real-time 60 FPS** performance within **256 KB** of memory by synchronizing operations with hardware interrupts and using **DMA** for data transfers.
- Designed **lifetime-aware memory allocators** using stack-based arenas and reusable free-list memory pools, providing predictable allocation on memory-constrained hardware.
- Built a Python CLI pipeline for asset compilation with deduplication, LZ77 compression, and content-based caching, reducing iteration time by **80%**.

Superscalar Out-of-Order RISC-V Processor | SystemVerilog, RISC-V, RTL Verification

Aug. 2025 – Nov. 2025

- Collaborated on a five-person team to design and implement a **3-way superscalar, out-of-order RV32IM processor** with register renaming, dynamic scheduling, speculative execution, and precise in-order retirement.
- Contributed to the integration of a **tournament branch predictor**, early branch recovery, banked instruction caching with prefetching and a victim cache, and a load-store queue with **byte-level store-to-load forwarding**.
- Verified the processor through module-level testbenches and automated comparisons against a **golden ISA simulator**, passing all provided benchmarks and achieving a synthesized **20 ns clock period**.

Approximate Distance Oracle Simulator | C++, Algorithms, Graph Theory, SDL2

Sept. 2024 – Dec. 2024

- Implemented a C++ **Thorup–Zwick distance oracle** with $O(k)$ query time and configurable tradeoffs among memory usage, preprocessing, and approximation accuracy.
- Built an interactive SDL2 visualization and validation environment, leading to the project's adoption as an experimentation tool in a **graduate-level algorithms course**.

Leadership & Involvement

Japanese Student Association

Sept. 2023 – May 2024

Club Finance Lead & Event Director

University of Michigan

- Planned and executed cross-cultural events in collaboration with executive board members, university departments, and student organizations, creating opportunities for cultural exchange and community engagement for **200+ attendees**.
- Coordinated event logistics, budgeting, and resource allocation with club leadership and campus partners, ensuring successful delivery of multiple large-scale programs while maintaining financial accountability.